

the bare-bones `real_sfs_bb.c`, along with the `real_sfs_ds.h` (based on the same file created earlier); these, and a supporting Makefile, are available from http://linuxforu.com/article_source_code/sept12/rsfbb_code.tbz2.

Bare-bones SFS module in action

Once compiled using `make`, getting the `real_sfs_bb.ko` driver, Pugs did his usual unusual experiments (see Figure 1). He checked the kernel window `/proc/filesystems` for kernel-supported file systems, then loaded the `real_sfs_bb.ko` driver and re-checked `/proc/filesystems` to find it showed `sfs` in the list. He mounted the pen drive partition `/dev/sdb1` onto `/mnt` using the `sfs` file system, checking the `dmesg` logs in the adjacent window. (Remember that right now, `sfs_fill_super` is not really reading the partition, and not doing any checks, so it really doesn't matter how `/dev/sdb1` is formatted.) But yes, the `mount` output shows that it is mounted using the `sfs` file system.

Oops! The `df` output, however, shows 'Function not implemented', and `cd` gives 'Not a directory'. Aha! Pugs hasn't yet implemented any functions in the other four function-pointer structures. That's still left to be done.

(Continued from Page 76....)

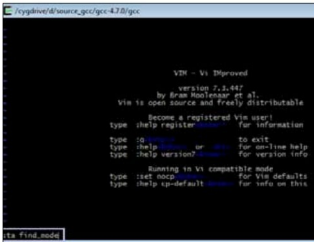


Figure 1: Executing the 'ta <tags>' command

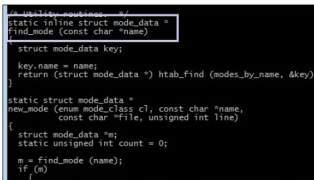


Figure 2: The definition of tag



Note: The above experiments use `sudo`, but one could just get into a root shell and do them without a `sudo`.

Okay, so no kernel crashes, and a bare-bones file system in action—Yippee!

Yes! Pugs knows that `df`, `cd`, ... are not yet functional. For that, he needs to start adding the various system calls in the other four function-pointer structures so that, like other file systems, people can use the various commands. And yes, Pugs is already onto his task—after all, he needs to have a geeky demo for his final semester project. 

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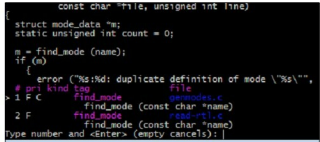


Figure 3: Executing the ':ts' command

compare it to Ctags. Till then, try using Ctags, and explore more on the Internet about it. 

References and further reading

- 1) http://vim.wikia.com/wiki/Browsing_programs_with_tags
- 2) <http://www.fsl.cs.sunysb.edu/~rick/cscope.html>

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